APPENDIX II—LIC DEBT SUSTAINABILITY ANALYSIS

INTERNATIONAL MONETARY FUND

TOGO

Debt Sustainability Analysis Under the Debt Sustainability Framework for Low-Income Countries

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May 24, 2010

In line with the joint World Bank-Fund low income country debt sustainability analysis (LIC DSA) produced with the third review of the Extended Credit Facility (ECF), this update of LIC DSA shows that Togo is in debt distress, demonstrating the need for reaching the completion point of the HIPC Initiative debt relief.[1] *An upward revision in growth, mainly due to strong agricultural production, is slightly improving the debt indicators, which offsets the impact of an ECF augmentation requested by the authorities. However, the key debt sustainability indicators are still above the relevant indicative thresholds for the next few years. An alternative scenario illustrating the impact of additional HIPC, MDRI and beyond-HIPC debt relief at the completion point suggests strong improvements in debt burden indicators. The inclusion of Togo's large domestic public debt in the analysis generally reinforces the conclusions of the external DSA.*

A. Background

1. **The last DSA for Togo prepared in 2009 concluded that Togo was in debt distress**. The outcome of this analysis is consistent with the previous DSAs finding improvement in debt dynamics as a consequence of expected enhancements in the macroeconomic policy framework, notably greater fiscal consolidation and solid implementation of growth-promoting structural reforms, large domestic arrears clearance

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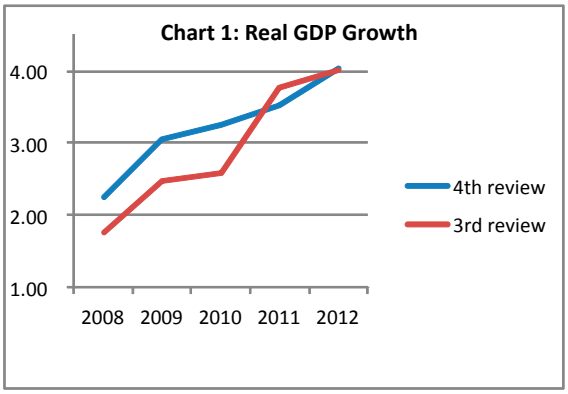
[1] This is a Fund staff update to the joint World Bank/IMF DSA presented with the report for the third review of the ECF (Country Report No. 09/173). See "Applying the Debt Sustainability Framework for Low-Income Countries Post Debt Relief" (IDA/SecM2006-0564, 8/11/06). Togo's quality of policies and institutions, as measured by the average World Bank's Country Policy and Institutional Assessment (CPIA) for the period 2006–08 (2.6), places it as a "weak performer". The corresponding indicative thresholds for the external debt indicators are 30 percent for the NPV of debt-to-GDP ratio, 100 percent of the debt-to-export ratio, 200 percent for NPV of debt-to-revenue ratio, 15 percent for the debt service-to-exports ratio and 25 percent for the debt service-to-revenue ratio.

operations and the HIPC initiative interim debt relief. The key indicative ratios for the current DSA are slightly better than in the 2009 DSA, mostly because the revised real GDP growth estimates from 2008 onward are stronger than projected during the previous exercise, mainly because of strong agricultural results[2].

2. **Since November 2008, Togo has been in the interim period of the HIPC Initiative.** Upon reaching the decision point, Togo was granted Cologne terms debt relief[3] by the Paris Club and negotiated a rescheduling agreement with several multilateral and non Paris Club creditors. Togo has signed bilateral agreements with most of its Paris Club creditors and has contacted the other non Paris Club creditors to seek comparable treatment as the one granted by the Paris Club. Also, the nominal debt stock fell from US$2.2 billion at end-2007 to US$1.7 billion at end-2008, reflecting the arrears clearance operations.

B. Baseline Assumptions

3. **Including a revision in growth rates, the baseline scenario stays consistent with the three-year ECF arrangement**. It is based on a continuation of steady growth averaging 3.7 percent from 2009 to 2019 and 4 percent from 2020 to 2029. The revised real GDP growth figures are higher from 2008 to 2012 (chart 1). The revision is due essentially to better-than-expected agricultural production (particularly in food production) and faster growth of the secondary sector (extractive industries, especially clinker). In addition, revisions to the GDP deflator for 2008 have caused a significant increase in nominal GDP compared to 3rd review estimates[4]. Under the assumptions of a stable political and social



[2] Agricultural production was particularly strong in 2009 (over 9 percent of real growth) partially because of favorable weather conditions and government policies to stimulate the sector such as the distribution of fertilizer at subsidized prices. Food production performed especially well. Among export crops, the increase in coffee production is notable.

[3] Cologne terms represent a 90 percent reduction of debt service falling due during the interim period and the remaining 10 percent is rescheduled with 6 year-grace period and 23 years of maturity for non ODA debt and 16 year-grace period and 40 years of maturity for ODA debt.

[4] The 2008 deflator was revised upwards relative to third review estimates. In particular, food prices and agricultural products prices increases (40 percent) were substantially higher compared to previous estimates, hence reflecting more fully the global food price increases observed for that year. Price increases for phosphate and clinker (18 percent) were also much higher than initially estimated.

situation, growth will be driven by several factors: donor-financed public investment, an improved investment climate, growing regional integration thanks to Togo's strategic geographical location and the role of the port of Lomé, a rebound in cotton and phosphate production following the restructuring of these sectors, and deeper financial intermediation after the ongoing restructuring of the banking sector.

4. **Macroeconomic stability will come from a prudent fiscal policy and structural reforms.** The domestic primary deficit is assumed to stay close to zero during the projected period. While the tax revenues are expected to increase with a broadening of the tax base, the resources freed up by HIPC and MDRI relief will be allocated to priority sectors such as health, education, and infrastructure.

Box 1. Togo-Macroeconomic Assumptions

The baseline macroeconomic framework assumes a stable political and social situation and that important structural reforms in the financial and real sector are brought to fruition, accelerating output growth to the estimated potential and attracting additional foreign direct investment over the medium-term.

- R**eal GDP growth** is projected to reach its estimated potential of 4 percent by 2013 and then vary around that level. Growth will be driven by the rehabilitation of the phosphate and cotton sectors, an improved investment climate, increased financial intermediation; additional FDI and foreign aid; and growing regional integration, thanks to Togo's strategic geographical location and the role of the port of Lomé.

- The projections for key **commodity prices** (oil, cotton, cocoa, and coffee) through 2015 are based on WEO projections of end May 2010 and are assumed constant in real terms afterwards.

- **Inflation** over the long term is projected to remain stable at 2 percent, reflecting sound monetary policy at the regional level.

- **The current account deficit** will remain roughly stable over the medium term, with higher exports of phosphates, cement and clinker being insufficient to compensate for a strong growth in imports as foreign aid is absorbed and foreign investment increases.

- **The domestic primary fiscal deficit** is assumed to stay close to zero during the projected period providing a fiscal anchor to ensure fiscal sustainability over the long term.

- **FDI and donor flows** are expected to increase over the medium term, reflecting improvements in the investment climate and overall governance.

5. **Under the assumption that FDI and donor flows increase over the medium term, the external position is projected to remain sustainable**. Togo's current account deficit is not expected to improve over the medium term. While exports are projected to pick up, largely on account of higher phosphate exports following the restructuring of the sector as well as continued growth in cement and clinker exports, imports are also expected to increase at a fast pace as foreign aid is absorbed and foreign investment increases. Sustained export growth will require enhancing competitiveness through reforms to improve the business environment.

6. **The baseline scenario reflects the multilateral arrears clearance operations completed in 2008 and assumes full delivery of traditional debt relief as well as interim HIPC assistance**.[5] Consistent with the DSF guidelines, the baseline does not reflect the delivery of HIPC, MDRI and bilateral or multilateral beyond-HIPC assistance after the completion point.[6]

C. External Debt Sustainability Analysis

Baseline

7. **Under the baseline scenario, Togo's external debt indicators remain above their relevant indicative thresholds demonstrating that the country is in debt distress** (Table 1a, Figure 1). The present value (PV) of public and publicly guaranteed (PPG) debt equals 41 percent of GDP in 2009 and remains above the 30 percent threshold until 2016. Both the PV of external debt relative to revenues and exports exceed their respective indicative threshold in 2014 emphasizing the need to reach the completion point and keeping a high degree of concessionality of Togo's future debt. However, the higher than projected level of agricultural and clinker production and the higher level of GDP compared to the third review, entail a higher level of exports, bringing the debt service ratios below their respective indicative thresholds for the whole projection period.

Alternative Scenarios and Stress Tests

8. **Togo's external debt outlook remains vulnerable to numerous shocks** (Table 1b, Figure 1). The PV of external debt to revenue and exports indicators deteriorate

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[5] Arrears to IDA and the AfDB were cleared in 2008. Togo has reached an agreement on arrears clearance with IFAD, OPEC and EIB. Togo is negotiating with BADEA. The IsDB has agreed in principle to provide relief but the modalities have not been defined yet. Togo has contacted FEGECE to ask for HIPC relief.

[6] See "Staff Guidance Note on the Application of the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries" (www.imf.org and IDA/SECM2007/0226, 03/05/2007).

significantly under a variety of shocks, in particular shocks that assume a depreciation of the exchange rate and lower export growth.

9. **Alternative scenarios assume additional delivery of debt relief at the completion point, which would significantly improve Togo's debt sustainability outlook** (Figure 2). Debt relief under the HIPC Initiative, MDRI and possible bilateral and multilateral beyond-HIPC assistance would significantly improve Togo's external debt outlook. Reaching the completion point, which is assumed to occur in 2010, and the resulting irrevocable debt relief would reduce all external debt indicators to levels below the relevant indicative thresholds.

D. Public Sector Debt Sustainability

Baseline

10. **The inclusion of Togo's large domestic public debt in the analysis emphasizes the vulnerability of the baseline scenario** (Table 2a, Figure 2). Togo's domestic debt burden is comparatively large, reflecting years of weak fiscal management and domestic arrears accumulation, as well as the need to recapitalize ailing banks. In addition, inclusion of the recent BCEAO credit linked to the SDR allocation worsens the key indicative ratios. However, this financing substitutes for planned issuance of securities which were not yet included and would have carried a higher debt service cost. The PV of total public debt is projected to remain relatively high during the next five year, hovering around 200 percent of revenues. Given the assumed improvement in the macroeconomic outlook and the projected high degree of concessionality of financing in the baseline scenario, debt ratios would nevertheless fall steadily over the long run.

Alternative scenarios and stress tests

11. **The evolution of the debt indicators would be sensitive to a variety of shocks, which would increase the debt level and debt service over the long run**. Total public debt dynamics are particularly vulnerable to a growth shock and to a lesser extent to a real depreciation (Table 2b, Figure 2). This highlights the importance of a reform agenda that improves the business environment to support foreign investment and growth.

12. **Full delivery of HIPC relief, MDRI, and beyond-HIPC relief at the assumed completion point reduces these vulnerabilities as shown in the alternative scenario.** All three PV-based indicators would be substantially lower than under the baseline and would decline further over the projection period.

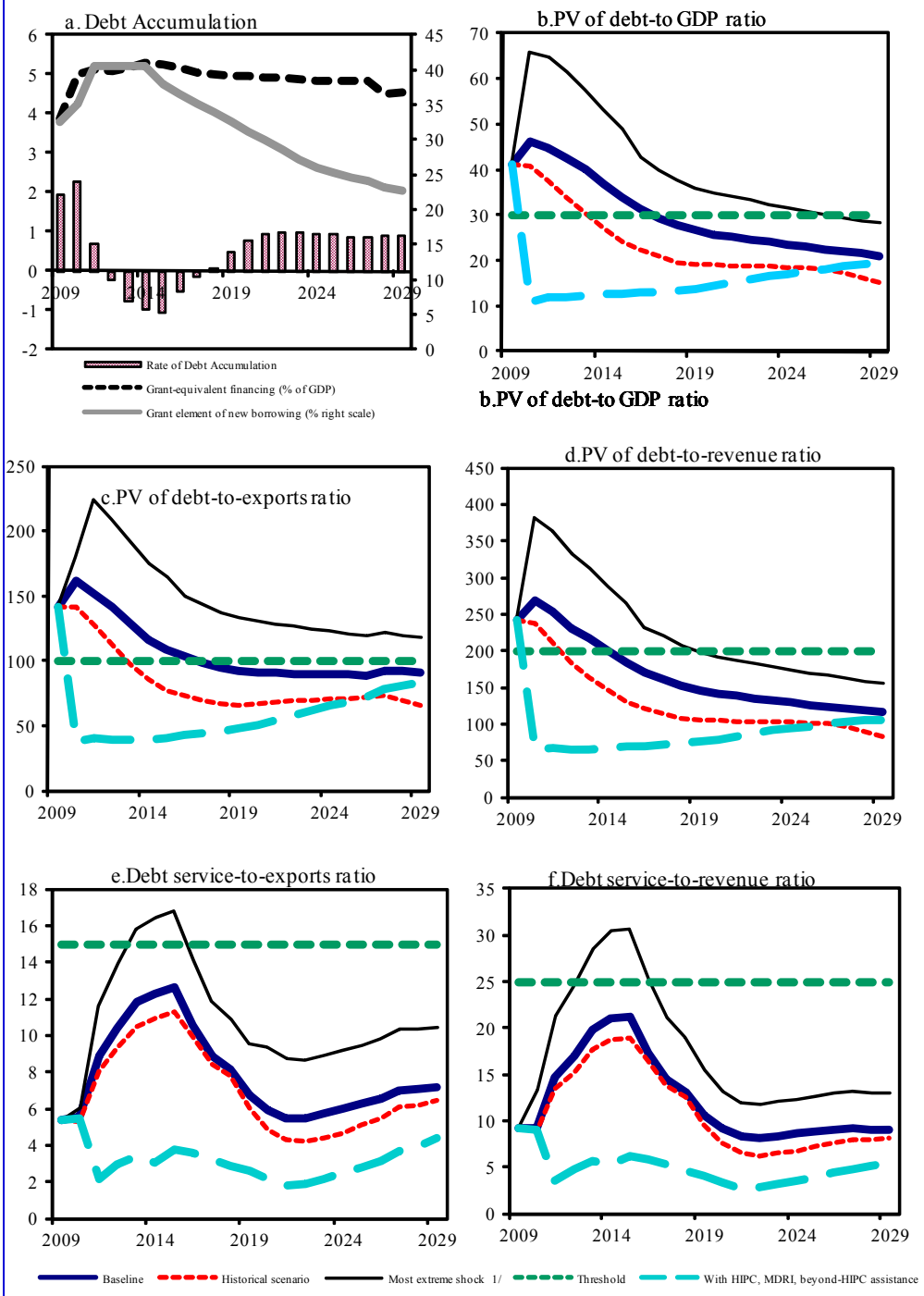
E. Conclusion

13. **The DSA shows that Togo remains in debt distress, despite significant debt relief already achieved under the HIPC initiative.** The authorities shared our view on the analysis and the conclusions of the rating. The debt reduction associated with the recent

external arrears clearance and by reaching the decision point of the HIPC Initiative does not suffice to allow a sustainable path of debt accumulation. Under the baseline scenario, the PV-based indicators remain well above their indicative thresholds for most of the projection period, emphasizing the importance for additional debt relief, which is expected upon reaching the completion point. Debt reduction upon reaching the HIPC completion point dramatically improves Togo's external debt outlook.

14. **Achieving a robust external debt outlook will depend on a sustained pick-up of real GDP growth, exports and foreign direct investment, as well as prudent debt management and solid fiscal performance.** Alternative scenarios and bound tests highlight the vulnerability of Togo's current external debt outlook. The inclusion of Togo's large domestic debt in the analysis reinforces the conclusions of the external DSA and emphasizes the risks to Togo's debt prospects. In this context, it is essential that the Togolese authorities continue current efforts to strengthen public finance management, restructure the banking system and promote financial development, reform state-owned enterprises, and improve the investment climate, laying the foundation for accelerating growth prospects.

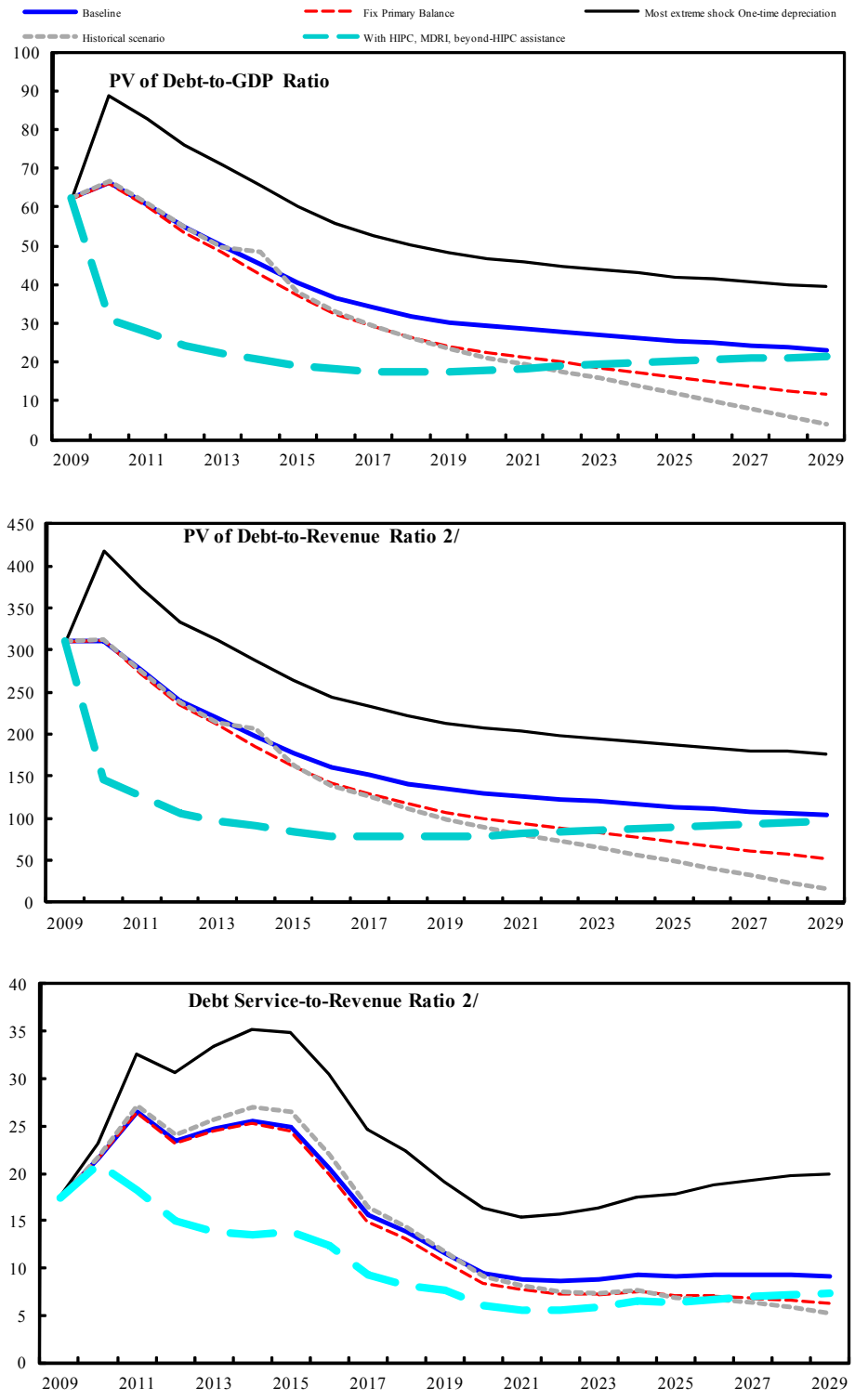
Figure 1. Togo: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2009-2029 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2019. In figure b. it corresponds to a With HIPC, MDRI, beyond-HIPC assistance shock; in c. to a With HIPC, MDRI, beyond-HIPC assistance shock; in d. to a With HIPC, MDRI, beyond-HIPC assistance shock; in e. to a With HIPC, MDRI, beyond-HIPC assistance shock and in figure f. to a With HIPC, MDRI, beyond-HIPC assistance shock

Figure 2.Togo: Indicators of Public Debt Under Alternative Scenarios, 2009-2029 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2019.
2/ Revenues are defined inclusive of grants.

Table 1a.: External Debt Sustainability Framework, Baseline Scenario, 2006-2029 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average	Standard Deviation	Projections																						
	2006	2007	2008			2009	2010	2011	2012	2013	2014	2009-2014 Average	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2015-2029 Average
External debt (nominal) 1/	**88.7**	**82.5**	**56.2**			**50.7**	**57.5**	**56.5**	**54.4**	**52.0**	**49.1**		**46.1**	**43.5**	**41.4**	**39.7**	**38.4**	**37.4**	**36.7**	**36.0**	**35.2**	**34.4**	**33.6**	**32.9**	**32.1**	**31.4**	**30.7**	
o/w public and publicly guaranteed (PPG)	88.7	82.5	55.6			49.5	55.6	54.0	51.5	48.6	45.3		42.1	39.5	37.2	35.3	33.9	32.9	32.1	31.3	30.5	29.8	29.0	28.3	27.6	27.0	26.3	
Change in external debt	-1.1	-6.2	-26.3			-5.4	6.8	-1.1	-2.0	-2.5	-2.9		-2.9	-2.6	-2.1	-1.7	-1.3	-0.9	-0.8	-0.7	-0.8	-0.8	-0.8	-0.7	-0.8	-0.7	-0.7	
Identified net debt-creating flows	-5.1	-8.7	-11.9			4.3	4.9	3.8	3.1	2.8	2.0		2.1	1.2	1.4	1.7	1.2	1.3	1.6	1.8	1.9	2.0	2.2	2.0	2.8	3.4	3.4	
Non-interest current account deficit	**1.1**	**4.3**	**5.9**	**4.5**	**5.2**	**6.7**	**7.2**	**5.8**	**5.2**	**5.1**	**4.6**		**5.2**	**4.2**	**4.4**	**4.6**	**4.6**	**4.7**	**5.0**	**5.2**	**5.3**	**5.4**	**5.5**	**5.6**	**6.6**	**7.1**	**7.1**	5.4
Deficit in balance of goods and services	12.7	14.8	14.4			14.6	16.3	17.1	16.6	16.0	15.2		15.7	14.3	14.3	14.3	14.3	14.3	14.3	14.3	14.3	14.3	14.3	14.3	15.2	15.2	15.2	
Exports	26.2	31.5	30.2			29.1	28.7	29.5	30.2	30.9	31.7		31.1	30.4	29.8	29.3	28.8	28.3	27.8	27.3	26.7	26.2	25.7	25.2	23.8	23.4	23.0	
Imports	38.9	46.3	44.6			43.7	45.0	46.6	46.8	47.0	46.9		46.7	44.7	44.2	43.6	43.1	42.6	42.1	41.5	41.0	40.5	40.0	39.6	39.0	38.6	38.2	
Net current transfers (negative = inflow)	-12.6	-11.4	-9.6	-10.2	2.5	-9.1	-9.7	-11.4	-11.2	-11.0	-10.8		-10.6	-10.4	-10.2	-10.0	-9.8	-9.7	-9.5	-9.3	-9.2	-9.0	-8.9	-8.7	-8.6	-8.2	-8.1	-9.3
o/w official	-1.4	-1.7	-1.4			-1.5	-2.5	-4.3	-4.3	-4.3	-4.3		-4.3	-4.3	-4.3	-4.3	-4.3	-4.3	-4.3	-4.3	-4.3	-4.3	-4.3	-4.3	-4.3	-4.0	-4.0	
Other current account flows (negative = net inflow)	1.1	0.9	1.1			1.2	0.6	0.1	-0.1	0.0	0.2		0.2	0.3	0.2	0.3	0.2	0.1	0.2	0.2	0.1	0.1	0.0	0.0	0.0	0.0	0.0	
Net FDI (negative = inflow)	**-4.1**	**-4.1**	**-1.3**	**-3.2**	**1.2**	**-1.0**	**-1.0**	**-1.0**	**-1.0**	**-1.0**	**-1.5**		**-2.0**	**-2.0**	**-2.0**	**-2.0**	**-2.5**	**-2.5**	**-2.5**	**-2.5**	**-2.5**	**-2.5**	**-2.5**	**-3.0**	**-3.0**	**-3.0**	**-3.0**	-2.5
Endogenous debt dynamics 2/	**-2.1**	**-8.8**	**-16.6**			**-1.4**	**-1.3**	**-1.1**	**-1.1**	**-1.3**	**-1.2**		**-1.1**	**-1.0**	**-1.0**	**-0.9**	**-0.9**	**-0.9**	**-0.8**	**-0.8**	**-0.8**	**-0.8**	**-0.8**	**-0.7**	**-0.8**	**-0.6**	**-0.7**	
Contribution from nominal interest rate	2.0	1.8	0.4			0.4	0.3	0.9	1.0	0.9	0.8		0.7	0.6	0.6	0.6	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	
Contribution from real GDP growth	-3.1	-1.5	-1.5			-1.7	-1.7	-2.0	-2.1	-2.1	-2.0		-1.9	-1.6	-1.6	-1.4	-1.4	-1.4	-1.4	-1.4	-1.4	-1.3	-1.3	-1.1	-1.3	-1.1	-1.2	
Contribution from price and exchange rate changes	-0.9	-9.2	-15.5			…	…	…	…	…	…		…	…	…	…	…	…	…	…	…	…	…	…	…	…	…	
Residual (3-4) 3/	**4.0**	**2.4**	**-14.4**			**-9.8**	**1.9**	**-4.8**	**-5.2**	**-5.3**	**-4.9**		**-5.0**	**-3.8**	**-3.5**	**-3.4**	**-2.5**	**-2.3**	**-2.4**	**-2.6**	**-2.7**	**-2.8**	**-3.0**	**-2.7**	**-3.6**	**-4.1**	**-4.1**	
o/w exceptional financing	-3.4	-2.5	29.8			0.3	22.2	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
PV of external debt 4/	...	...	45.1			42.4	48.1	47.3	45.6	43.4	40.7		37.9	35.5	33.6	32.1	31.1	30.3	29.8	29.3	28.7	28.2	27.6	27.1	26.5	26.0	25.5	
In percent of exports	...	...	149.4			145.6	168.0	160.6	151.1	140.2	128.4		122.1	116.7	112.7	109.5	107.7	107.1	107.2	107.4	107.5	107.6	107.5	107.4	111.0	110.9	110.8	
PV of PPG external debt	**...**	**...**	**44.4**			**41.1**	**46.2**	**44.9**	**42.7**	**40.0**	**37.0**		**33.9**	**31.4**	**29.4**	**27.7**	**26.6**	**25.8**	**25.1**	**24.6**	**24.1**	**23.5**	**23.0**	**22.5**	**21.9**	**21.5**	**21.1**	
In percent of exports	**...**	**...**	**147.3**			**141.2**	**161.4**	**152.3**	**141.4**	**129.3**	**116.6**		**109.1**	**103.4**	**98.5**	**94.6**	**92.2**	**91.0**	**90.5**	**90.3**	**90.1**	**89.9**	**89.6**	**89.2**	**92.0**	**91.8**	**91.7**	
In percent of government revenues	**...**	**...**	**285.1**			**241.4**	**269.3**	**252.1**	**230.7**	**216.3**	**199.8**		**183.3**	**169.9**	**161.6**	**152.4**	**146.0**	**141.5**	**138.2**	**135.2**	**132.2**	**129.3**	**126.3**	**123.7**	**120.5**	**118.2**	**115.8**	
Debt service-to-exports ratio (in percent)	**13.6**	**11.2**	**7.3**			**5.4**	**5.5**	**8.8**	**10.4**	**11.8**	**12.3**		**12.6**	**10.6**	**8.9**	**8.1**	**6.7**	**5.9**	**5.4**	**5.5**	**5.7**	**6.0**	**6.2**	**6.5**	**7.0**	**7.1**	**7.2**	
PPG debt service-to-exports ratio (in percent)	**13.6**	**11.2**	**7.3**			**5.4**	**5.5**	**8.8**	**10.4**	**11.8**	**12.3**		**12.6**	**10.6**	**8.9**	**8.1**	**6.7**	**5.9**	**5.4**	**5.5**	**5.7**	**6.0**	**6.2**	**6.5**	**7.0**	**7.1**	**7.2**	
PPG debt service-to-revenue ratio (in percent)	**21.1**	**21.0**	**14.0**			**9.1**	**9.2**	**14.6**	**16.9**	**19.8**	**21.1**		**21.2**	**17.4**	**14.5**	**13.1**	**10.7**	**9.1**	**8.3**	**8.2**	**8.4**	**8.6**	**8.8**	**9.1**	**9.2**	**9.1**	**9.1**	
Total gross financing need (Billions of U.S. dollars)	0.0	0.1	0.2			0.2	0.3	0.3	0.3	0.3	0.3		0.4	0.3	0.3	0.3	0.3	0.3	0.4	0.4	0.4	0.5	0.5	0.5	0.7	0.7	0.8	
Non-interest current account deficit that stabilizes debt ratio	2.2	10.6	32.2			12.1	0.4	6.9	7.3	7.5	7.5		8.2	6.8	6.5	6.3	5.9	5.6	5.7	5.9	6.0	6.2	6.3	6.3	7.5	7.7	7.8	
Key macroeconomic assumptions																												
Real GDP growth (in percent)	3.7	1.9	2.2	1.6	2.2	3.1	3.3	3.5	3.9	4.1	4.0	3.6	4.0	3.8	3.9	3.7	3.8	3.8	3.9	3.9	4.0	4.1	4.1	3.6	4.2	3.7	4.0	3.9
GDP deflator in US dollar terms (change in percent)	1.1	11.5	23.2	6.4	10.0	-3.7	-3.2	-0.4	0.7	0.6	1.4	-0.8	1.8	8.1	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.5	2.4	2.5	2.4	2.8
Effective interest rate (percent) 5/	2.3	2.3	0.6	1.9	0.7	0.7	0.7	1.6	1.8	1.7	1.6	1.3	1.5	1.5	1.4	1.4	1.4	1.4	1.5	1.5	1.5	1.5	1.5	1.5	1.6	1.6	1.6	1.5
Growth of exports of G&S (US dollar terms, in percent)	-16.8	36.4	20.6	9.1	19.3	-4.3	-1.5	6.1	7.1	7.3	8.0	3.8	3.8	9.9	4.4	4.4	4.4	4.4	4.4	4.4	4.4	4.5	4.5	4.5	0.8	4.5	4.5	4.5
Growth of imports of G&S (US dollar terms, in percent)	-1.8	35.1	21.3	9.2	17.7	-2.8	2.9	6.8	5.0	5.2	5.3	3.7	5.4	7.4	5.1	4.9	5.0	5.0	5.1	5.2	5.2	5.2	5.2	5.2	5.3	5.2	5.2	5.3
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	32.7	35.2	40.6	40.6	40.6	40.6	38.4	37.9	36.6	35.2	33.9	32.6	31.2	29.9	28.6	27.2	25.9	25.4	24.6	24.0	23.2	22.7	29.3
Government revenues (excluding grants, in percent of GDP)	16.9	16.9	15.6			17.0	17.2	17.8	18.5	18.5	18.5		18.5	18.5	18.2	18.2	18.2	18.2	18.2	18.2	18.2	18.2	18.2	18.2	18.2	18.2	18.2	18.2
Aid flows (in Billions of US dollars) 7/	0.0	0.1	0.1			0.1	0.2	0.2	0.2	0.2	0.3		0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.4	0.4	0.4	0.4	0.4	0.5	0.4	0.5	
o/w Grants	0.0	0.0	0.1			0.1	0.1	0.1	0.1	0.2	0.2		0.2	0.2	0.2	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.4	0.4	0.4	0.4	
o/w Concessional loans	0.0	0.0	0.0			0.0	0.1	0.1	0.1	0.1	0.1		0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			3.9	5.0	5.1	5.1	5.2	5.3		5.2	5.2	5.0	5.0	5.0	4.9	4.9	4.9	4.9	4.9	4.8	4.8	4.8	4.5	4.5	4.9
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			68.1	74.3	81.2	81.3	80.0	78.4		77.4	77.2	78.6	78.4	78.2	77.6	77.1	76.6	76.0	75.4	75.2	74.8	74.5	72.9	72.9	76.2
Memorandum items:																												
Nominal GDP (Billions of US dollars)	2.2	2.5	3.2			3.2	3.1	3.2	3.4	3.6	3.7		4.0	4.5	4.7	5.0	5.4	5.7	6.1	6.4	6.9	7.3	7.8	8.3	8.8	9.4	10.0	
Nominal dollar GDP growth	4.8	13.6	25.9			-0.8	0.0	3.1	4.6	4.7	5.4	2.8	5.9	12.3	6.4	6.2	6.3	6.3	6.4	6.5	6.5	6.6	6.6	6.2	6.7	6.3	6.6	6.8
PV of PPG external debt (in Billions of US dollars)			1.3			1.4	1.4	1.4	1.4	1.4	1.4		1.3	1.3	1.3	1.3	1.3	1.4	1.4	1.5	1.6	1.6	1.7	1.8	1.8	1.9	2.0	
(PVt-PVt-1)/GDPt-1 (in percent)						2.0	2.3	0.7	-0.2	-0.8	-1.0	0.5	-1.0	-0.5	-0.2	0.1	0.5	0.8	0.9	1.0	1.0	1.0	0.9	0.9	0.9	0.9	0.9	0.5

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt, does not include the US$ 32 million supplier credit signed in [ ].

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
The large amount in 2008 reflects the arrears clearance operation.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1b.Togo: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2009-2029
(In percent)

											Projections										
	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	**2019**	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of debt-to GDP ratio																					
Baseline	41	46	45	43	40	37	34	31	29	28	**27**	26	25	25	24	24	23	23	22	22	21
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2009-2029 1/	41	41	37	34	30	27	24	22	21	19	**19**	19	19	19	19	19	18	18	17	16	15
A2. New public sector loans on less favorable terms in 2009-2029 2	41	46	45	43	41	39	37	33	31	30	**30**	29	29	29	29	28	28	28	28	28	28
A3. Alternative Scenario :full delivery of HIPC, MDRI, and beyond HIPC debt relief	43	11	12	12	12	13	13	13	13	13	**14**	14	15	16	17	17	18	18	19	19	19
B. Bound Tests																					
B1. Real GDP growth at historical average minus one standard deviation in 2010-2011	41	47	48	46	43	40	37	32	30	28	**27**	26	25	25	24	24	23	23	22	22	21
B2. Export value growth at historical average minus one standard deviation in 2010-2011 3/	41	47	51	49	46	43	40	35	33	31	**30**	29	28	27	26	25	24	23	22	22	21
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010-2011	41	46	48	45	43	39	36	32	30	28	**27**	26	25	25	24	23	23	22	22	21	21
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010-2011 4/	41	45	46	44	41	38	35	31	29	27	**26**	25	24	24	23	22	22	21	21	20	20
B5. Combination of B1-B4 using one-half standard deviation shocks	41	44	47	44	42	39	35	31	29	27	**26**	25	25	24	24	23	22	22	21	21	20
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	41	66	65	61	58	53	49	43	40	38	**36**	35	34	33	32	32	31	30	29	29	28
PV of debt-to-exports ratio																					
Baseline	141	161	152	141	129	117	109	103	99	95	**92**	91	91	90	90	90	90	89	92	92	92
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2009-2029 1/	141	142	127	112	98	85	77	73	70	66	**66**	67	68	69	70	71	71	72	73	70	66
A2. New public sector loans on less favorable terms in 2009-2029 2	141	159	153	144	134	123	118	108	105	103	**102**	103	104	106	107	108	110	111	117	118	120
A3. Alternative Scenario :full delivery of HIPC, MDRI, and beyond HIPC debt relief	141	38	40	40	39	40	41	42	44	46	**48**	51	54	58	62	65	69	72	78	81	84
B. Bound Tests																					
B1. Real GDP growth at historical average minus one standard deviation in 2010-2011	141	158	151	140	128	116	109	97	93	89	**86**	85	84	84	84	83	83	83	85	85	85
B2. Export value growth at historical average minus one standard deviation in 2010-2011 3/	141	180	224	209	192	175	165	149	143	138	**134**	131	128	126	125	123	121	119	121	120	118
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010-2011	141	158	151	140	128	116	109	97	93	89	**86**	85	84	84	84	83	83	83	85	85	85
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010-2011 4/	141	158	156	144	132	120	112	101	96	92	**90**	88	87	87	86	86	85	84	87	86	86
B5. Combination of B1-B4 using one-half standard deviation shocks	141	153	162	150	137	124	116	104	99	95	**93**	91	91	90	90	89	89	88	91	90	90
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	141	158	151	140	128	116	109	97	93	89	**86**	85	84	84	84	83	83	83	85	85	85
PV of debt-to-revenue ratio																					
Baseline	241	269	252	231	216	200	183	170	162	152	**146**	142	138	135	132	129	126	124	121	118	116
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2009-2029 1/	241	237	210	184	163	147	130	121	114	107	**105**	104	104	104	103	102	100	100	95	90	84
A2. New public sector loans on less favorable terms in 2009-2029 2	241	265	253	235	224	211	198	178	172	166	**162**	160	159	158	157	156	155	154	153	152	152
A3. Alternative Scenario :full delivery of HIPC, MDRI, and beyond HIPC debt relief	241	64	66	65	66	68	69	70	72	74	**76**	79	83	87	91	94	97	100	102	104	106
B. Bound Tests																					
B1. Real GDP growth at historical average minus one standard deviation in 2010-2011	241	275	272	248	233	215	198	173	164	155	**148**	143	140	136	133	130	127	124	121	118	116
B2. Export value growth at historical average minus one standard deviation in 2010-2011 3/	241	274	286	263	248	232	215	189	181	171	**164**	157	151	146	141	137	132	127	123	119	115
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010-2011	241	269	268	245	230	213	195	171	162	153	**146**	142	138	135	132	129	125	123	119	117	114
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010-2011 4/	241	264	258	236	221	205	189	166	157	148	**142**	137	133	130	127	123	120	117	114	111	108
B5. Combination of B1-B4 using one-half standard deviation shocks	241	258	263	241	226	209	192	168	160	151	**144**	140	136	133	129	126	123	120	117	114	112
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	241	383	363	332	311	288	264	232	220	207	**198**	192	187	182	178	174	170	166	161	158	155

Table 1b.Togo: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2009-2029 (continued)
(In percent)

Debt service-to-exports ratio

Baseline	5	6	9	10	12	12	13	11	9	8	**7**	6	5	5	6	6	6	7	7	7	7
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2009-2029 1/	5	5	8	9	11	11	11	10	8	8	**6**	5	4	4	4	5	5	6	6	6	6
A2. New public sector loans on less favorable terms in 2009-2029 2	5	6	9	10	12	13	12	10	9	8	**7**	7	7	7	7	7	8	8	9	9	9
A3. Alternative Scenario :full delivery of HIPC, MDRI, and beyond HIPC debt relief	5	5	2	3	3	3	4	4	3	3	**3**	2	2	2	2	2	3	3	4	4	4
B. Bound Tests																					
B1. Real GDP growth at historical average minus one standard deviation in 2010-2011	5	6	9	10	12	12	13	11	9	8	**7**	6	5	5	6	6	6	6	7	7	7
B2. Export value growth at historical average minus one standard deviation in 2010-2011 3/	5	6	12	14	16	16	17	14	12	11	**10**	9	9	9	9	9	9	10	10	10	10
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010-2011	5	6	9	10	12	12	13	11	9	8	**7**	6	5	5	6	6	6	6	7	7	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010-2011 4/	5	6	9	10	12	12	13	11	9	8	**7**	6	6	6	6	6	6	7	7	7	7
B5. Combination of B1-B4 using one-half standard deviation shocks	5	5	9	11	13	13	13	11	9	9	**7**	6	6	6	6	6	7	7	7	7	8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	5	6	9	10	12	12	13	11	9	8	**7**	6	5	5	6	6	6	6	7	7	7

Debt service-to-revenue ratio

Baseline	9	9	15	17	20	21	21	17	15	13	**11**	9	8	8	8	9	9	9	9	9	9
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2009-2029 1/	9	9	13	15	18	19	19	16	14	13	**9**	8	7	6	6	7	7	8	8	8	8
A2. New public sector loans on less favorable terms in 2009-2029 2	9	9	15	17	20	22	21	16	14	13	**11**	10	10	10	11	11	11	12	12	12	12
A3. Alternative Scenario :full delivery of HIPC, MDRI, and beyond HIPC debt relief	9	9	4	5	6	5	6	6	5	5	4	3	3	3	3	4	4	4	5	5	6
B. Bound Tests																					
B1. Real GDP growth at historical average minus one standard deviation in 2010-2011	9	10	16	18	21	23	23	19	16	14	**12**	10	9	9	9	9	9	10	10	10	10
B2. Export value growth at historical average minus one standard deviation in 2010-2011 3/	9	9	15	18	20	22	22	18	15	14	**12**	11	10	10	10	10	10	10	11	10	10
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010-2011	9	9	16	18	21	23	23	19	16	14	**11**	10	9	9	9	9	9	10	10	10	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010-2011 4/	9	9	15	17	20	21	21	17	15	13	**11**	10	9	9	9	9	9	9	9	9	9
B5. Combination of B1-B4 using one-half standard deviation shocks	9	9	15	18	21	22	22	18	15	14	**11**	10	9	9	9	9	9	9	10	9	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	9	13	21	24	29	31	31	25	21	19	**15**	13	12	12	12	12	13	13	13	13	13
Memorandum item:																					
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	31	31	31	31	31	31	31	31	31	31	**31**	31	31	31	31	31	31	31	31	31	31

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a. Togo: Public Sector Debt Sustainability Framework, Baseline Scenario, 2006-2029
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2006	2007	2008	Average	Standard Deviation	2009	2010	2011	2012	2013	2014	2009-14 Average	2019	2029	2015-29 Average
Public sector debt 1/	115.1	109.1	83.9			70.6	75.6	70.0	63.5	58.6	53.5		37.6	28.4	
o/w foreign-currency denominated	88.7	82.5	55.6			49.5	55.6	54.0	51.5	48.6	45.3		33.9	26.3	
Change in public sector debt	2.0	-6.1	-25.1			-13.4	5.0	-5.6	-6.5	-4.9	-5.1		-1.8	-0.7	
Identified debt-creating flows	-9.5	-10.6	-26.7			-6.8	3.7	-0.9	-1.2	-1.0	-1.2		-0.8	-0.3	
Primary deficit	1.7	-0.3	-0.5	-0.3	2.0	0.3	0.3	1.2	0.8	0.9	1.1	0.8	1.0	1.1	1.1
Revenue and grants	18.3	18.5	17.8			20.0	21.2	22.1	22.8	22.8	22.8		22.5	22.2	
of which: grants	1.4	1.7	2.2			3.0	4.1	4.3	4.3	4.3	4.3		4.3	4.0	
Primary (noninterest) expenditure	20.0	18.3	17.2			20.3	21.5	23.3	23.6	23.7	23.9		23.5	23.3	
Automatic debt dynamics	-11.1	-10.3	-9.9			-6.8	3.5	-2.1	-2.0	-1.9	-2.3		-1.8	-1.3	
Contribution from interest rate/growth differential	-2.0	-2.3	-15.5			-2.8	-2.4	-2.7	-2.7	-2.5	-2.7		-1.8	-1.3	
of which: contribution from average real interest rate	2.0	-0.1	-13.1			-0.3	-0.2	-0.1	-0.1	0.0	-0.4		-0.4	-0.2	
of which: contribution from real GDP growth	-4.0	-2.1	-2.4			-2.5	-2.2	-2.6	-2.6	-2.5	-2.2		-1.4	-1.1	
Contribution from real exchange rate depreciation	-9.1	-8.0	5.5			-4.0	5.9	0.5	0.7	0.6	0.4		...	...	
Other identified debt-creating flows	-0.1	0.0	-16.2			-0.3	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	-0.1	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	-16.2			-0.3	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	11.4	4.5	1.6			-6.6	1.3	-4.6	-5.3	-3.9	-4.0		-1.1	-0.4	
Other Sustainability Indicators															
PV of public sector debt	26.4	26.6	72.8			62.2	66.2	60.9	54.7	50.0	45.1		30.3	23.2	
o/w foreign-currency denominated	0.0	0.0	44.4			41.1	46.2	44.9	42.7	40.0	37.0		26.6	21.1	
o/w external	...	...	44.4			41.1	46.2	44.9	42.7	40.0	37.0		26.6	21.1	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	6.0	4.0	2.9			3.8	4.9	7.1	6.1	6.6	6.9		3.7	3.1	
PV of public sector debt-to-revenue and grants ratio (in percent)	144.4	143.3	409.8			310.6	311.7	275.6	239.8	219.3	197.9		134.5	104.1	
PV of public sector debt-to-revenue ratio (in percent)	156.2	157.7	467.1			365.4	385.5	342.1	295.6	270.3	243.8		166.3	127.2	
o/w external 3/	…	…	285.1			241.4	269.3	252.1	230.7	216.3	199.8		146.0	115.8	
Debt service-to-revenue and grants ratio (in percent) 4/	23.2	23.1	19.3			17.5	21.6	26.5	23.3	24.7	25.6		11.6	9.1	
Debt service-to-revenue ratio (in percent) 4/	25.1	25.4	22.0			20.5	26.7	32.9	28.8	30.4	31.5		14.4	11.2	
Primary deficit that stabilizes the debt-to-GDP ratio	-0.2	5.8	24.6			13.7	-4.7	6.8	7.3	5.8	6.2		2.9	1.8	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	3.7	1.9	2.2	1.6	2.2	3.1	3.3	3.5	3.9	4.1	4.0	3.6	3.8	4.0	3.9
Average nominal interest rate on forex debt (in percent)	2.3	2.3	0.6	1.9	0.7	0.7	0.7	1.6	1.8	1.8	1.7	1.4	1.5	1.7	1.6
Average nominal interest rate on domestic debt (in percent)	0.8	1.1	0.9	1.0	0.2	2.6	3.4	2.5	2.1	2.1	1.9	2.4	0.6	0.9	1.0
Average real interest rate (in percent)	1.8	-0.1	-12.3	-1.1	6.5	-0.3	-0.3	-0.1	-0.1	0.0	-0.7	-0.3	-1.0	-0.7	-0.9
Average real interest rate on foreign-currency debt (in percent)	-0.2	-2.1	-12.8	-2.8	6.4	-1.6	-1.7	-1.9	-1.9	-1.7	-2.4	-1.9	-2.3	-2.3	-2.3
Average real interest rate on domestic debt (in percent)	0.6	-1.0	-12.1	-4.1	6.9	0.9	1.6	0.5	0.1	0.3	-0.5	0.5	-1.8	-1.5	-1.4
Real exchange rate depreciation (in percent, + indicates depreciation)	-10.3	-9.2	7.8	-4.4	10.6	-7.6	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	0.2	2.1	14.8	3.3	6.6	1.7	1.8	2.0	2.0	1.8	2.5	2.0	2.4	2.4	2.4
Growth of real primary spending (deflated by GDP deflator, in percent)	0.1	-0.1	0.0	0.0	0.2	0.2	0.1	0.1	0.1	0.0	0.0	0.1	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	…	…	32.7	35.2	40.6	40.6	40.6	40.6	38.4	32.6	22.7	...

Sources: Country authorities; and staff estimates and projections.

1/ Covers public and publicly guaranteed debt including state-owned enterprises debt. Does not include the US$ 32 million supplier credit signed in [ ].

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2b.Togo: Sensitivity Analysis for Key Indicators of Public Debt 2009-2029

	Projections							
	2009	2010	2011	2012	2013	2014	2019	2029
PV of Debt-to-GDP Ratio								
Baseline	62	66	61	55	50	45	30	23
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	62	67	61	55	50	48	23	4
A2. Primary balance is unchanged from 2009	62	66	60	53	48	43	24	12
A3. Permanently lower GDP growth 1/	62	67	62	56	52	47	36	42
A4. Alternative Scenario :With HIPC, MDRI, Beyond-HIPC assistance	61	31	31	31	31	31	31	31
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2010-2011	62	69	68	63	59	56	46	49
B2. Primary balance is at historical average minus one standard deviations in 2010-2011	62	68	63	57	52	47	32	25
B3. Combination of B1-B2 using one half standard deviation shocks	62	68	64	59	55	51	40	41
B4. One-time 30 percent real depreciation in 2010	62	89	83	76	71	65	48	39
B5. 10 percent of GDP increase in other debt-creating flows in 2010	62	76	71	65	60	55	39	30
PV of Debt-to-Revenue Ratio 2/								
Baseline	311	312	276	240	219	198	134	104
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	311	313	274	237	214	207	99	16
A2. Primary balance is unchanged from 2009	311	312	272	234	211	186	107	52
A3. Permanently lower GDP growth 1/	311	313	279	245	226	207	160	185
A4. Alternative Scenario :With HIPC, MDRI, Beyond-HIPC assistance	311	146	126	105	97	91	78	96
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2010-2011	311	325	304	272	257	240	202	218
B2. Primary balance is at historical average minus one standard deviations in 2010-2011	311	318	284	248	228	206	142	110
B3. Combination of B1-B2 using one half standard deviation shocks	311	321	288	256	239	221	176	181
B4. One-time 30 percent real depreciation in 2010	311	417	374	334	311	287	214	177
B5. 10 percent of GDP increase in other debt-creating flows in 2010	311	359	321	283	263	241	174	136
Debt Service-to-Revenue Ratio 2/								
Baseline	17	22	26	23	25	26	12	9
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	17	22	27	24	26	27	12	5
A2. Primary balance is unchanged from 2009	17	22	26	23	24	25	11	6
A3. Permanently lower GDP growth 1/	17	22	27	24	25	26	13	13
A4. Alternative Scenario :With HIPC, MDRI, Beyond-HIPC assistance	17	21	18	15	14	14	8	7
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2010-2011	17	22	28	25	27	28	14	16
B2. Primary balance is at historical average minus one standard deviations in 2010-2011	17	22	27	24	25	26	12	10
B3. Combination of B1-B2 using one half standard deviation shocks	17	22	28	24	26	27	13	14
B4. One-time 30 percent real depreciation in 2010	17	23	33	31	33	35	19	20
B5. 10 percent of GDP increase in other debt-creating flows in 2010	17	22	28	26	26	27	13	12

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.